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Snowflake Assignment-4 (Analyze Query Performance Using Cache)

Step 1: Executed Query for the First Time

Query - 01c4d1c3-0001-e862-001c-9f4b001351ee

TEST_WAREHOUSE MIHIR

Analyze with Cortex Code

Query Details

Query Profile

Query Telemetry

Details

Status

Success

Duration

940ms

Driver Status

N/A

Start Time

6/4/2026, 1:01:05 PM

Query ID

01c4d1c3-0001-e862-001c-9f4b001351ee

Client Driver

Snowsight

End Time

6/4/2026, 1:01:06 PM

Query Tag

Session ID

8056443820224574

Warehouse Size

X-Small

Attributed Compute Credits

Parameterized query hash

116ef2c8f7b776ead2a2326b7836c110

SQL Text

SELECT SUM(L_EXTENDEDPRICE)
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM
WHERE L_DISCOUNT > 0.05;

Query - 01c4d1c3-0001-e862-001c-9f4b001351ee

TEST_WAREHOUSE MIHIR

Analyze with Cortex Code

Query Details

Query Profile

Query Telemetry

Result [0]

0%

SUM(LINEITEM.L_EXTENDEDPRICE)

1

Aggregate [1]

0%

SUM(LINEITEM.L_EXTENDEDPRICE)

2.727M

Filter [2]

0%

LINEITEM.L_DISCOUNT > 0.05

2.727M

TableScan [3]

95.8%

SNOWFLAKE_SAMPLE_DATA.TPCH_SF1...

Query insights

None detected. Learn more

Most expensive nodes(1 of 4)

TableScan [3] 95.8%

Profile overview(Finished)

Total execution time (807ms) 100.0%

Processing 1.4%

Remote disk I/O 38.1%

Synchronization 58.3%

Initialization 4.2%

Statistics

Scan progress 100.00%

Bytes scanned 21.84MB

Percentage scanned from cache 0.00%

Partitions scanned 10

Partitions total 10

Showing that Table Scan is Done, thus taking 940ms.

Step 2: Execute the query again immediately after the first run.

Query - 01c4d1c4-0001-e940-001c-9f4b0012f966

TEST_WAREHOUSE MIHIR

Analyze with Cortex Code

Query Details

Query Profile

Query Telemetry

Details

Status

Success

Duration

204ms

Driver Status

N/A

Start Time

6/4/2026, 1:02:53 PM

Query ID

01c4d1c4-0001-e940-001c-9f4b0012f966

Client Driver

Snowsight

End Time

6/4/2026, 1:02:53 PM

Query Tag

—

Session ID

8056443820224574

Warehouse Size

X-Small

Attributed Compute Credits

—

Parameterized query hash

116ef2c8f7b776ead2a2326b7836c110

SQL Text

SELECT SUM(L_EXTENDEDPRICE)
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM
WHERE L_DISCOUNT > 0.05;

Query - 01c4d1c4-0001-e940-001c-9f4b0012f966

TEST_WAREHOUSE MIHIR

Analyze with Cortex Code

Query Details

Query Profile

Query Telemetry

Query insights

None detected. Learn more

Profile overview(Finished)

Total execution time

(86ms) 100.0%

Initialization

100.0%

Statistics

Scan progress

100.00%

Bytes scanned

20.55MB

Percentage scanned from cache

100.00%

Partitions scanned

10

Partitions total

10

Result [0]

0%

SUM(LINEITEM.L_EXTENDEDPRICE)

1

Aggregate [1]

0%

SUM(LINEITEM.L_EXTENDEDPRICE)

2.727M

Filter [2]

0%

LINEITEM.L_DISCOUNT > 0.05

2.727M

TableScan [3]

0%

SNOWFLAKE_SAMPLE_DATA.TPCH_SF1...

Showing that Table Scan is not Done and instead CACHE is Used, thus taking 204ms.

Step 3: Created a new query that reads from the same table but accesses different columns.

Query - 01c4d1ce-0001-e9cc-001c-9f4b001370f2

TEST_WAREHOUSE MIHIR

Analyze with Cortex Code

Query Details

Query Profile

Query Telemetry

Details

Status

Success

Duration

464ms

Driver Status

N/A

Start Time

6/4/2026, 1:12:28 PM

Query ID

01c4d1ce-0001-e9cc-001c-9f4b001370f2

Client Driver

Snowsight

End Time

6/4/2026, 1:12:29 PM

Query Tag

—

Session ID

8056443820224574

Warehouse Size

X-Small

Attributed Compute Credits

—

Parameterized query hash

3dc57a9e4f6bd62700d29e995455aef

SQL Text

```
select count(L_QUANTITY)
from SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM
where L_QUANTITY > 40;
```

Query - 01c4d1ce-0001-e9cc-001c-9f4b001370f2

TEST_WAREHOUSE MIHIR

Analyze with Cortex Code

Query Details

Query Profile

Query Telemetry

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Result [0]

COUNT(LINEITEM.L_QUANTITY)

0%

1

Aggregate [1]

COUNT(LINEITEM.L_QUANTITY)

0%

1.2M

Filter [2]

LINEITEM.L_QUANTITY > 40

0%

1.2M

TableScan [3]

SNOWFLAKE_SAMPLE_DATA.TPCH_SF1...

0%

Query insights

None detected. [Learn more](#)

Profile overview(Finished)

Total execution time

(132ms) 100.0%

Initialization

100.0%

Statistics

Scan progress

100.00%

Bytes scanned

4.68MB

Percentage scanned from cache

100.00%

Partitions scanned

10

Partitions total

10

Conclusion :- Snowflake leverages metadata cache to optimize query parsing and planning.